
Department of Civil Engineering, IIT Madras
CV2090 (Probability & Statistics for Civil Engineers)
Jul–Nov 2026

03 Aug, 2026

Tutorial-1

3:00–5:00 pm

Instructions:

1. Make assumptions wherever necessary, and *state* them explicitly.
 2. Tutorials are closed-phone, closed-tablet, and closed-LLM.
 3. You can bring books and notes, you are welcome to discuss with friends and TAs. However, your submission should be your own work.
 4. Submit handwritten solutions on A4 sheets, prepared during the tutorial hours.
 5. Submissions will not be accepted after the tutorial hours.
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Q1. Autonomous Vehicle Multi-Sensor Perception An autonomous vehicle utilizes three primary perception sensors: LiDAR (L), Radar (R), and an Optical Camera (C). During operation under adverse weather conditions, each sensor is classified as either operational (1) or failed (0).

- (a) Define the sample space Ω as an explicit set of binary triplets (L, R, C) .
- (b) Express Event A , “At least two sensors are operational,” and Event B , “The LiDAR sensor has failed,” as explicit subsets of Ω .
- (c) Compute the set $A \cap B^c$.

Q2. Cryptographic Hash Collisions in IoT Networks An Internet of Things (IoT) edge device uses a lightweight cryptographic hashing algorithm that outputs hash values into an 8-bit output space ($N = 2^8 = 256$ distinct hash buckets).

- (a) Assuming the hash function distributes inputs uniformly and independently across all buckets, write the analytical expression for the probability $\Pr(C)$ that among $k = 10$ randomly generated sensor signatures, at least two signatures produce identical hash values (a hash collision).
- (b) Evaluate the probability correct to three significant figures of at least one collision $\Pr(C)$.

Q3. Conditional Probabilities in Structural Fatigue A civil engineering bridge monitoring framework employs a primary acoustic strain sensor (S_1) and a secondary optical displacement sensor (S_2). The baseline probability that S_1 triggers a fatigue alert is $\Pr(S_1) = 0.05$. If S_1 triggers, the conditional probability that S_2 also triggers is $\Pr(S_2|S_1) = 0.60$. If S_1 does not trigger, the probability that S_2 triggers is $\Pr(S_2|S_1^c) = 0.02$.

- (a) Express the event that *neither sensor triggers an alert* using set operations.
- (b) Calculate $\Pr(S_1^c \cap S_2^c)$ strictly using probability axioms and conditional probability rules.

Q4. Union and Complement In a startup pitch competition, 30% of startups secure seed funding (event A) and 50% secure venture capital (event B). If 20% get both seed

funding and venture capital ($A \cap B$), find $\Pr(A \cup B)$, the probability a startup secures at least one form of funding.

Q5. Inclusion–Exclusion (Three Events). In a tech lab, engineers may know Python (event P), Java (event J), and C++ (event C). Suppose $\Pr(P) = 0.7$, $\Pr(J) = 0.6$, $\Pr(C) = 0.5$, with $\Pr(P \cap J) = 0.5$, $\Pr(J \cap C) = 0.4$, $\Pr(P \cap C) = 0.3$, $\Pr(P \cap J \cap C) = 0.2$. Find the probability an engineer knows at least one of these languages $\Pr(P \cup J \cup C)$.

Q6. Total Probability (Partition). In a city, 60% of cars are manually driven and 40% are autonomous. Manual cars have an accident rate of 3%, and autonomous cars 5%. Using the law of total probability, what is the overall accident rate for a randomly chosen car?

Q7. Bayes' Theorem (Spam Filter). A spam filter is 99% sensitive (detects spam when present) and 98% specific (does not flag non-spam as spam). Suppose 1% of emails are actually spam. If an email is flagged as spam, what is the probability it truly is spam? (Apply Bayes' theorem.)

- (a) Calculate the probability of total mission success, $\Pr(S_1 \cap S_2 \cap S_3)$.
- (b) Given that a mission failed ($F = (S_1 \cap S_2 \cap S_3)^c$), calculate the conditional probability that the failure occurred specifically during Second-Stage Staging (meaning Stage 1 succeeded, but Stage 2 failed).

Q8. Pairwise vs. Mutual Independence A quantum error-correction system generates three binary register states $X_1, X_2, X_3 \in \{0, 1\}$. Independent physical processes generate two fair classical random bits $B_1, B_2 \in \{0, 1\}$, where $\Pr(B_1 = 1) = 0.5$ and $\Pr(B_2 = 1) = 0.5$. The quantum registers are assigned as $X_1 = B_1$, $X_2 = B_2$, and $X_3 = B_1 \oplus B_2$ (where $\oplus$ denotes **addition modulo 2**). Define events $A = \{X_1 = 1\}$, $B = \{X_2 = 1\}$, and $C = \{X_3 = 1\}$,

- (a) Prove that events A , B , and C are pairwise independent.
- (b) Prove that events A , B , and C are not mutually independent by evaluating $\Pr(A \cap B \cap C)$ relative to $\Pr(A) \Pr(B) \Pr(C)$.
- (c) What are the implications of above findings?

Q9. Simpson's Paradox (Data Reversal). Two treatments, Drug A and Drug B, are being compared for treating a Cancer. Patients belong to one of two groups:

- Group 1 (Mild disease): Recovery rates are 90% on Drug A and 80% on Drug B.
- Group 2 (Severe disease): Recovery rates are 30% on Drug A and 20% on Drug B.

Thus, Drug A has a higher recovery rate than Drug B in both groups. However, the treatments were prescribed to different numbers of patients:

- Drug A was given to 20 patients in Group 1 and 100 patients in Group 2.
 - Drug B was given to 100 patients in Group 1 and 20 patients in Group 2.
- (a) Compute the overall recovery rate for Treatment A.
 - (b) Compute the overall recovery rate for Treatment B.
 - (c) Explain how Drug B can have a lower recovery than A within each group but a higher overall recovery (**Simpson's paradox**).